



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2016
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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H2 BIOLOGY

9648/03

Applications Paper and Planning Question

20 September 2016

Paper 3

2 hours

Additional Materials: Answer papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough workings.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

At the end of the examination,

1. Fasten all your work securely together.
2. Hand in the following separately:
 - Structured questions 1 – 3
 - Planning question
 - Essay question

The number of marks is given in brackets [] at the end of each question or part question.

ANSWER SCHEME

For examiner's Use	
1	/ 13
2	/ 13
3	/ 14
4 Planning	/ 12
5 Essay	/ 20
Total	/ 72

This paper consists of __ printed pages.

[Turn over]

Answer **all** questions.

QUESTION 1

ABL1 is a proto-oncogene that encodes a protein tyrosine kinase involved in a variety of cellular processes, including cell division. A researcher intended to mass produce *ABL1* protein tyrosine kinase using bacterial cells. He obtained *ABL1* cDNA from the cDNA library and the complete sequence of *ABL1* cDNA non-template strand is shown in **Fig. 1.1**. The start and stop triplets are bolded.

```

5' 1 CCTATTATTACTTTATGGGGCAGCAGCCTGGATCCGT-----
61 -----
121 -----GCATCTGACTTTG
181 AGCCTCAGGG TCTGAGTGAAGCTTCT 3'
  
```

Fig. 1.1

- (a) Explain why the gene of interest is obtained from cDNA library instead of genomic DNA library. [2]
- cDNA library contains the *ABL1* gene without introns while genomic DNA library contains *ABL1* gene with introns
 - Thus *ABL1* can be successfully expressed/ used to synthesise functional protein by bacteria.

The plasmid pUC18, was first chosen for the cloning of *ABL1* cDNA. **Fig. 1.2a** shows a diagram of the plasmid pUC18 and the position of the restriction sites found in the plasmid.

Fig. 1.2b shows two restriction sites commonly used in genetic engineering.

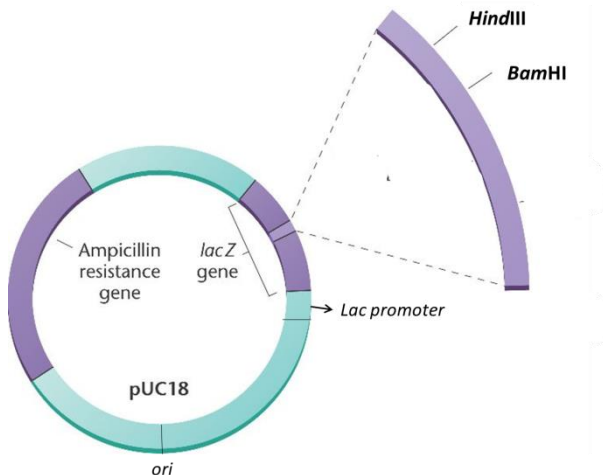


Fig. 1.2a

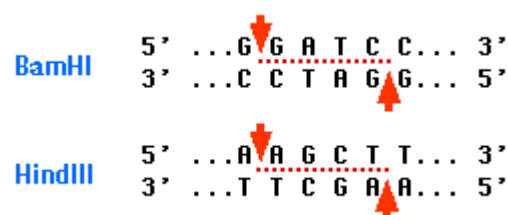


Fig. 1.2b

- (b) With reference to **Fig. 1.1** and **Fig. 1.2**, comment on the effectiveness of using pUC18 for *ABL1* protein production. [3]
- **Ineffective**
 - The **BamHI** restriction site is downstream of the start codon in *ABL1* cDNA (OWTTE) and **HindIII** restriction site is downstream of stop codon
 - Hence the cDNA cloned into the plasmid will be incomplete (OWTTE) and non-functional proteins will be produced.

The researcher found that the *ABL1* cDNA is present in low copy number in the cDNA library. Hence, he carried out PCR to amplify the *ABL1* cDNA.

(c) Suggest two reasons why the total amount of amplified *ABL1* cDNA did not increase between the 30th cycle and the 40th cycle. [2]

- Nucleotides are used up
- Primers are used up
- *Taq* DNA polymerase gradually becomes denatured

An alternative cloning technique called **TA cloning** shown in **Fig. 1.3** can also be used to clone the *ABL1* cDNA. The *Taq* DNA polymerase used in PCR has a non-template dependent activity which preferentially adds a single adenine nucleotide to the 3'-ends of a double stranded DNA molecule. This results in PCR product with 3'-A overhangs. This enables the *ABL1* cDNA to be inserted into a plasmid designed to have 3'-T overhangs.

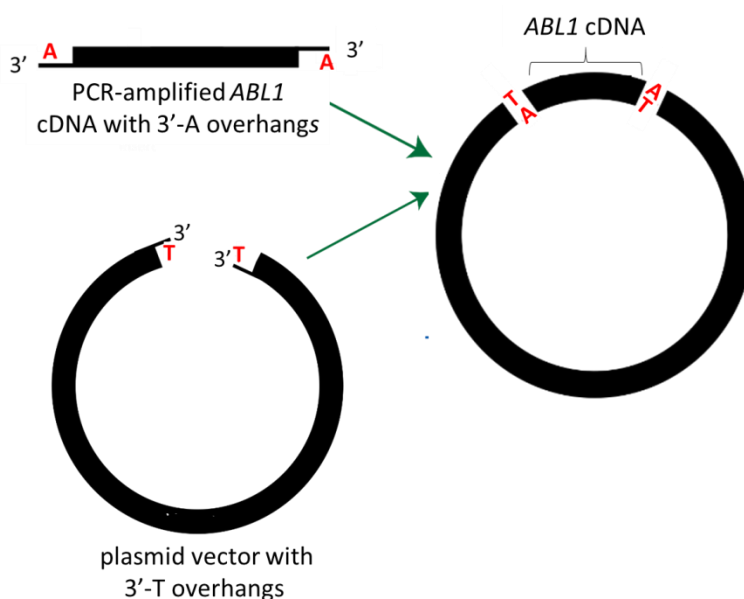


Fig. 1.3

(d) (i) With reference to **Fig. 1.3**, describe how the PCR-amplified *ABL1* cDNA can be inserted into the plasmid using **TA cloning** technique. [4]

- PCR products of *ABL1* cDNA contain 3'-A overhangs and the plasmid have 3'-T overhang
- Mix the PCR products with the plasmid
- Anneal by complementary base pairing between A and T through formation of hydrogen bonds
- DNA ligase seals the sugar-phosphate backbone by forming phosphodiester bonds

(ii) Suggest one advantage and one disadvantage of using **TA cloning**. [2]

Advantage:

- Allows cloning of GOI that does not have restriction sites / do not require the use of restriction enzymes
- Prevents reannealing of plasmid via complementary base pairing

Disadvantage:

- Does not allow directional insertion of GOI (OWTTE), hence bacteria transformed with recombinant plasmid may produce wrong/non-functional proteins.

[Total: 13]

QUESTION 2

Plants have developed defence mechanisms against pathogens such as bacteria, fungi and viruses. Chemicals released by these pathogens can trigger a defence response in infected plant cells. For example, the production of hydrogen peroxide (H_2O_2) which reacts with pathogen membranes and cellular chemicals eventually kills both the cell and pathogen.

The *OSRac1* gene from another plant species was isolated and introduced into a number of rice plant (*Oryza spp.*) lines to study its role in disease resistance of plants to blast fungus. Experiments were carried out to see if the *OSRac1* gene was part of the signalling pathway for hydrogen peroxide production. A control (C) and four other genetically modified rice plant lines (A1, A2, D1 and D2) grown *in vitro* from calluses were exposed to chemicals known to initiate a defence response by producing hydrogen peroxide. A1 and A2 are rice plants with the *OSRac1* gene always turned on. D1 and D2 are rice plants with the *OSRac1* gene suppressed. The results are shown in the Fig. 2.1.

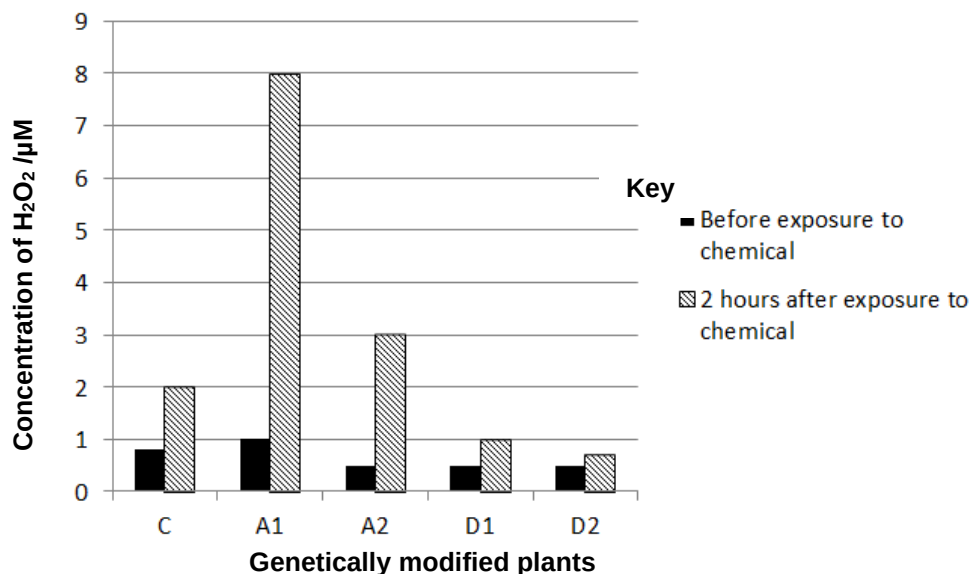


Fig. 2.1

(a) Describe how calluses are obtained from rice plants. [3]

- Surface of **explants** from rice is **sterilized** using dilute **sodium hypochlorite**.
- Rice explants are **aseptically** transferred to **sterile** culture vessels **containing nutrients** and **intermediate auxin:cytokinin ratio**
- ...to **induce callus formation** by **mitosis**.

(b) With reference to Fig. 2.1, compare the **change in H_2O_2 production** between the control and genetically modified plants two hours after the chemical was applied. [2]

- A1 and A2 showed a **higher increase** in H_2O_2 production of **8 times and 6 times respectively** as compared to **control** which showed an increase in H_2O_2 production of **2.5 times**
OR
A1 and A2 showed a **higher increase** in H_2O_2 production **by 700% and 500% respectively** as compared to **control** which showed an increase in H_2O_2 production of **150%**
- D1 and D2 which showed a **lower increase in H_2O_2 production of 2.0 and 1.4 times** as compared to **control** which showed an increase in H_2O_2 production of **2.5 times**

OR

- **D1 and D2** which showed a lower increase in H₂O₂ production by 100% and 40% as compared to **control** which showed an increase in H₂O₂ production of 150%

(c) Evaluate whether the data supports the hypothesis that *OSRac1* gene is involved in disease resistance. [2]

- **Supported** [1] *[mark awarded only if full explanation is given]*
 - **Both A1 and A2 genetically modified plants** with *OSRac1* gene always turned on showed greater change in the number of times of H₂O₂ production, so hypothesis is supported.
- OR
- **Both D1 and D2 genetically modified plants** with *OSRac1* gene suppressed showed smaller change in the number of times of H₂O₂ production, so hypothesis is supported.

Rice blast caused by the fungal pathogen is a destructive disease of rice. The use of blast resistance genes is an effective way to control the fungal disease in rice and to reduce losses in crop yield.

Recently, researchers identified a known genetic marker that is tightly linked to the blast resistance genes in some fungal resistant crops. This allows the identification of crops with blast resistance and subsequent cloning of transgenic rice with the blast resistance gene.

In a rice breeding programme, researchers wanted to identify the blast resistant crops from those that are susceptible to blast.

(d) Using the information provided, describe how RFLP analysis can help to distinguish between blast resistant crops and those that are susceptible to blast. [6]

1. **DNA samples of blast resistant and susceptible crops are cut with the same specific restriction enzyme to generate DNA fragments of different lengths**
2. **Restriction fragments are subjected to gel electrophoresis where the fragments are separated according to molecular weight/size**
10. **larger DNA fragments travel slower and smaller fragments travel faster.**
3. DNA being **negatively charged** due to the presence of phosphate group, moves towards the **anode** / positive end of the electric field.
4. A **DNA ladder** is loaded in another well to calibrate the **size of DNA fragments**.
5. **Add bromophenol blue and glycerol with purpose given**
6. (a) Southern blot and Nucleic acid hybridization are carried out .. **sodium hydroxide** denature double-stranded DNA to **single-stranded DNA**...
- (b) ...using a **radioactively-labelled DNA/RNA probe complementary to the tightly linked RFLP marker**
7. DNA bands of interest are detected using **autoradiography**

8. **Analyze and compare** the unique band patterns of DNA of the blast resistant and susceptible crops.
9. **[idea of ..]** The RFLP variant that is linked to the resistant allele will give a different band pattern to the RFLP variant that is linked to the susceptible allele.

[Total: 13]

QUESTION 3

(a) Describe how the normal copy of a gene can be introduced to a patient's cells via non-viral method. [3]

Mention any **one** of the methods:

Liposome method

- **Cationic liposomes** made of positively charged lipids are created.
- Positively charged liposomes are attracted to negatively charged recombinant plasmid DNA, forming liposome-DNA complexes.
- The liposome-DNA complexes are then **introduced directly into the target cells via endocytosis**.

Cationic polymer

- **Cationic polymers** designed to be able to **bind to the receptors** on the **target cells** are used.
- **Positively-charged** cationic polymer is attracted to **negatively-charged** recombinant DNA plasmid, forming a polymer-DNA complex/ polyplexes.
- The polyplexes bind to receptors on the target cells in culture, hence allowing the **uptake** of the complex by receptor mediated endocytosis forming endosomes.

Direct Injection and Electroporation

- The **normal allele** is inserted into a **DNA plasmid** to form a recombinant plasmid DNA.
- The recombinant plasmid DNA is injected directly into the tissue /**in vivo** technique
- The target cells are subjected to electrical current which **increases** the **permeability** of the membrane for the cells to take up DNA plasmids.

Gene gun

- The **normal allele** is inserted into a **DNA plasmid** to form a recombinant plasmid DNA.
- The recombinant plasmid DNA is coated onto microscopic gold / tungsten particles
- A **gene gun** is used to accelerate these particles towards the target cells in culture/ ex vivo
- These particles **penetrate** the **plasma membrane** of the target cells and deliver the DNA into the **nucleus**

Type 1 diabetes is a condition where an individual's Islet cells cannot produce insulin in response to high blood glucose levels. Patients usually are dependent on insulin injections.

Transplanting Islet cells from donors has been studied as a form of treatment for Type I diabetes for over three decades.

Islet cells are usually encapsulated in alginate microspheres before transplanting them into patients. The alginate microsphere creates a barrier between the donor cells and the recipient's cells.

Fig. 3.1 shows some Islet cells encapsulated in an alginate microsphere.

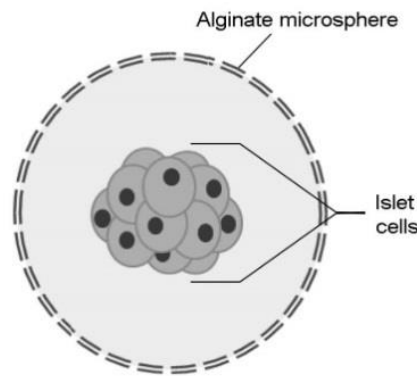


Fig. 3.1

(b) Suggest why Islet cells were encapsulated before they were transplanted into a patient. [1]

- Idea of immune system rejection.

Patients transplanted with human Islet cells obtained from deceased individuals can be made insulin independent for around 5 years using the Islet encapsulation treatment. However, this approach is limited because of the scarcity and quality of donor Islet cells.

(c) Suggest why Islet encapsulation treatment lasts only approximately 5 years. [2]

- The donated islet cells are **specialised** cells that **cannot divide / renew themselves**.
- Hence the islets cells would **die** and **stop producing insulin** after 5 years, hence the treatment would stop being effective.

OR

- The alginate microsphere **degrades** after 5 years,
- ...the patient's immune system destroys the islet cells, hence no more insulin is produced.

- (d) Researchers have also looked into producing human insulin-producing beta cells from stem cells.

A recent study in 2014 used a human embryonic stem cell line (HUES8) and a human induced pluripotent stem cell (hiPSC) line to develop two types of human insulin-producing beta cells called HUES8 SC- β and hiPSC SC- β respectively to overcome the problem of scarcity.

The human induced pluripotent stem cells were derived from fully differentiated adult somatic cells as shown in **Fig. 3.2**.

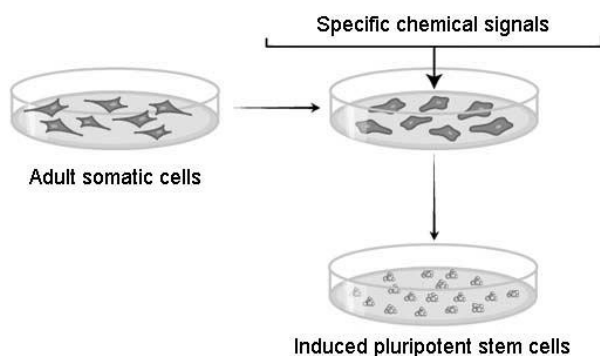


Fig. 3.2

- (i) Explain why fully differentiated somatic cells can be induced to become pluripotent. [2]

- Fully differentiated somatic cells contain the **entire genome**
- **Chemical signals** enable the cells to **express genes** important for maintaining **pluripotency** / idea that **genes for pluripotency are turned on**.

- (ii) State one ethical issue related to stem cell research and explain how using induced pluripotent stem cells would address this issue. [2]

- Pluripotent cells are usually obtained by **removing the inner cell mass of a blastocyst** and this is problematic as some believe that **life begins at conception** and this is **equivalent to destroying a human life**.
- Using iPSCs does not have as many ethical issues as **no embryos were destroyed in the process**.

- (e) Trials have been done by transplanting HUES8 SC- β cells and hiPSC SC- β cells into diabetic mice using alginate microsphere encapsulation. Human insulin produced by the mice was measured following a high carbohydrate meal.

The results of this study were recorded in **Table 3.1**.

Type of beta cell transplanted into diabetic mice	Mean concentration of human insulin secreted $\pm$ standard deviation / μg per ml of blood
HUES8 SC- β cells	2.3 ± 0.2
hiPSC SC- β cells	2.2 ± 0.3
Normal human beta cells	2.1 ± 0.9

Table 3.1

- (i) Compare the secretion of insulin by these three types of transplanted cells. [3]

Similarity:

- High glucose concentrations resulted in **all three types of cells secreting approximately the same amount of insulin / relatively similar mean amount of insulin**.
- Normal human beta cells, HUES8 SC- β cells and hiPSC SC- β cells secreted 2.1 2.3, 2.2 μg of insulin per ml of blood respectively.

Difference:

- HUES8 SC- β cells and hiPSC SC- β cells showed smaller variation in insulin secretion (with standard deviation of 0.2 μg of insulin per ml of blood and 0.3 μg of insulin per ml of blood respectively) than normal human beta cells (0.9 μg of insulin per ml of blood).

- (ii) Explain the purpose of the normal human beta cells. [1]

- As a control / reference to ascertain if HUES8 SC- β cells and hiPSC SC- β cells could produce comparable levels **[key idea]** of insulin in response to high glucose concentrations.

[Total: 14]

QUESTION 4 – Planning question

You are required to plan, but not carry out, an experiment to investigate the effect of temperature on the rate of respiration in mung beans.

Germination of mung beans can be initiated by soaking the seeds overnight. The mung beans can then be placed into a simple respirometer (**Fig. 4.1**) to measure the rate of respiration by measuring oxygen uptake of the seeds. Soda lime pellets absorb any carbon dioxide produced by the germinating seeds. As oxygen is taken up during respiration, the drop of coloured liquid introduced in the capillary tube by capillary action is displaced.

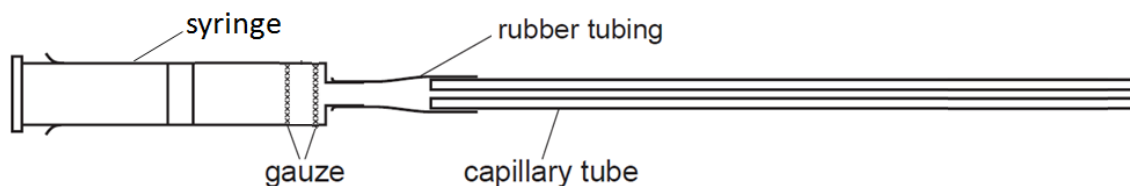


Fig. 4.1: A respirometer

You must use the items from this list:

- 200 mung beans of equal size that have been soaked for 24 hours
- Soda lime pellets
- Syringes
- Rubber tubing connected to glass capillary tube with 1mm bore diameter
- Beaker of coloured liquid
- Ruler marked in mm
- Paper towels
- Stop watch
- Thermostatically controlled incubator
- Other available laboratory apparatus and equipment



Your plan should have a clear and helpful structure to include:

- a description of the method used including the scientific reasoning behind the method,
- an explanation of the dependent and independent variables involved,
- relevant, clearly labelled diagrams,
- how you will record your results and ensure that they are as accurate and as reliable as possible,
- proposed layout of results tables and graphs with clear headings and labels,
- the correct use of technical and scientific terms,
- relevant risks and precautions taken

[Total: 12]

Suggested answer scheme:

Linking theory to investigation:

[1]

- O₂ acts as the **final electron acceptor** in the **electron transport chain** during **oxidative phosphorylation**. The protons and electrons combine with oxygen to form water. The higher the rate of respiration, the higher the rate of O₂ uptake, and the further the displacement of the coloured drop
- As temperature increases, kinetic energy increases, more enzyme-substrate complexes are formed, hence the rate of respiration increases. At optimum temperature, rate of respiration is highest. When temperatures goes beyond the optimum temperature, 3D structure of enzymes involved in respiration disrupted, enzymes are denatured, rate of respiration decreases.

Stating the hypothesis

[1]

- As temperature **increases towards the optimum**, rate of **oxygen consumption will increase**, reflecting an **increase in the rate of respiration**.

Independent and dependent variables

[1, both variables]

Independent variable: Temperature / °C, with 5 values within a reasonable range (e.g. 15°C, 25°C °C, 35°C, 45°C, 55°C) maintained using an incubator.

Dependent variable: Distance moved by the drop of coloured liquid in 5 min / mm

Variables to be kept constant and scientific reasoning [1, min 1 variable and rationale]

- **Duration** of each experiment should be kept constant to ensure a fair comparison.
- **Number** of **germinating mung beans** used - determines the concentration of enzymes present, and hence affects the rate of respiration. Hence, use the same number of beans (less than or equal to 10 per experiment) for each temperature.
- **Mass of soda lime pellets** (in excess) used should be kept constant as carbon dioxide production will affect the reading of the volume of oxygen absorbed / used up / distance moved by drop of liquid.
- **Volume of air in syringe** used should be kept constant as the volume will affect the amount of oxygen available for respiration.

Methods [2]

- Setting up the respirometer – putting in mung beans, soda lime pellets, coloured liquid
 - Describing method to measure dependent variable
 - How other variables are kept constant – number of seeds, duration in incubator
 - Equilibration of set-up to each temperature / **acclimatisation**
1. Remove the plunger from the syringe and place 2g of soda lime inside the syringe.
 2. Weigh 5 g of the germinating mung beans (*Accept 10 or less mung beans*) using a weighing balance and place them in the syringe barrel and replace the plunger by pushing it in until it is about 0.5 cm from the germinating seeds. Connect the glass capillary tube securely to the syringe via the rubber connecting tubing.
 3. Dip the end of the glass capillary tube into the coloured liquid so that a drop enters the capillary tube. Remove any excess liquid with paper towels.

- Set the incubator to 15°C and place the respirometer horizontally on the incubator shelf.
- Place a ruler beside the capillary tube and mark the position of the coloured drop.
- Wait for 3 minutes to ensure equilibration/acclimatization of temperature between the respirometer and the incubator. Also ensure that the drop of coloured liquid is moving smoothly towards the syringe.
- After 5 minutes, record the distance travelled by the drop of coloured liquid.

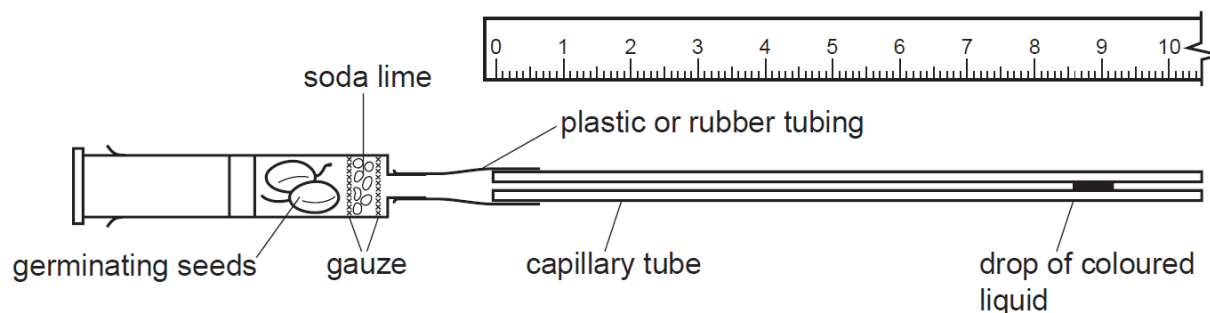
Replicates and repeats [1 for 8-9]

- Repeat steps 1 to 7 to obtain another two readings for this temperature to ensure **accuracy by taking the average reading**.
- Repeat steps 1 to 8 for the other four temperatures, using fresh germinating seeds.
- Repeat the experiment at **least two more times to ensure reproducibility / reliability** of experimental results.

Controls and justification [1]

- Replace germinating beans with equal mass of **boiled beans** at the optimum temperature.
- Rationale: To show that oxygen uptake is due to the germinating beans undergoing respiration and no other factors

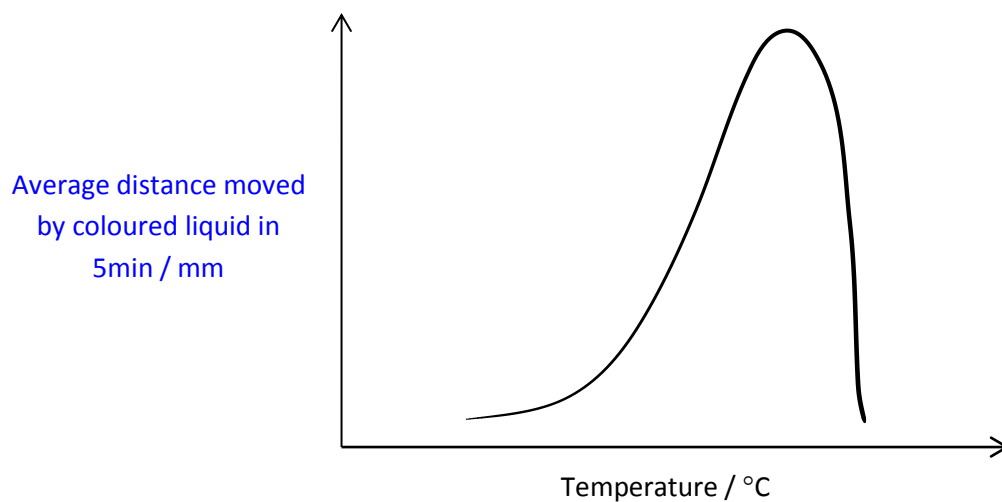
Labelled diagram [1]



Proposed layout of table with appropriate headings (independent & dependent variables consistent with axes of graph) [1]

Temperature / °C	Distance travelled by coloured liquid in 5 minutes / mm			
	Reading 1	Reading 2	Reading 3	Average
15				
25				
35				
45				
55				

Theoretical graph with correct axes and units (optimum temperature included) [1]



Risk assessment and precautions taken [1]

- Soda lime is an irritant/corrosive; handle with gloves and wash affected area immediately if it comes into contact with skin.

QUESTION 5

- (a) Discuss the detrimental environmental and economic effects of growing genetically-modified herbicide resistant crops. [6]

Environmental effects:

- (1) **Genetically-modified crop plants** may be hardier and become agricultural weeds that invade natural habitats
- (2) The **introduced transgene(s) to wild species** may result in more invasive hybrid offspring when pollen transfer to wild relatives.
- (3) **Intensive** use of herbicide selects for herbicide-resistant weeds
- (4) Intensive use of herbicide **reduces biodiversity, upsetting the natural balance of the ecological system.**

Economic effects (4 max):

- (5) High cost of GM seeds/plants, farmers cannot afford/ erode farmers' income
- (6) Heavy use of herbicide, thus, cost more
- (7) contamination of organic farming due to accidental mixing of GM crops with non-GM crops
- (8) cleaning pollution associated with heavy use of herbicide
- (9) human health problems associated with the use of herbicide

- (b) Discuss the ethical and social issues of the Human Genome Project. [6]

Ethical issues arise...

- (1) on the fairness in the use of genetic information by insurers or other organisations, etc which may create a situation in which less healthy individuals / those with genetic predisposition are marginalised or discriminated.
- (2) on the confidentiality and privacy of genetic information - the right of ownership of the genetic information has yet to be determined.
- (3) on the use of genetic information in controversial decisions in reproduction, such as modifying genotypes through gene therapy. Hence resulting in the ethical issue of
 - a) one generation determining the next generation's genotype before birth without their consent.
- OR
- b) going against nature by tampering with the next generation's genetic make-up.
- (4) on the use of genetic information in controversial decisions in reproduction, such as termination of pregnancies. Issue – selective breeding / abortion based on genetic tests that may not be fully reliable/understood. → increased abortion rate / decline in appreciation of the dignity of life
- (5) from clinical issues that the doctors, other health service providers, patients and the public must be educated to make informed choices, and be aware of scientific capabilities and limitations.

OR

Unethical for doctors or health care providers to carry out genetic tests

- a) and draw conclusions on data that may not be fully reliable, **or**
- b) without informing the patient of the limitations and capabilities of genetic technology, **or**
- c) leaving patient to interpret complex results without sufficient explanation.

- (6) from commercialization/patenting of genetic information and their product by companies which may limit their accessibility and development of useful medicinal products.
- (7) from the conceptual and philosophical understanding of human responsibility, free will versus genetic determinism, and concepts of health and disease. E.g. difficult to determine if a person's behaviour is genetically determined or can be controlled by the individual.
- (8) when an individual identified to have life-threatening genetic condition experience immense psychological ramification.

Social issue arises

- (9) Due to possibility of social and economic stigmatization/discrimination as well as genetic discrimination that arises due to the knowledge of individual's genetic difference.
- (c) Describe how a genetic condition like SCID may be treated with viral gene therapy and discuss the potential limitations of this kind of treatment. [8]

SCID gene therapy: Outline **[Max 6]**

- (1) SCID (both types accepted) is a **recessive** condition that may be treated by the introduction of the **dominant normal allele**.
- (2) A **retrovirus vector** is **modified** such that it **does not cause disease**.
- (3) **Hematopoietic stem cells or T cells** are harvested from the **SCID patient** and are cultured *ex vivo*.
- (4) **RNA** copies of **normal human ADA gene** or **ILRG-2 gene** are obtained from **bone marrow** of a donor.
- (5) This **ADA RNA** is then used to make a **recombinant RNA** molecule, which is **packaged into the modified retrovirus vector**.
- (6) The retrovirus is allowed to **infect** the harvested hematopoietic stem cells / T cells. The recombinant RNA molecules are **injected** into the cells.
- (7) **Reverse transcription** catalysed by viral reverse transcriptase occurs and **double-stranded complementary copies of ADA DNA** are produced.
- (8) The normal ADA allele **integrates randomly** into the host **genome**. Expression of the ADA allele produces functional ADA enzymes.
- (9) The **hematopoietic stem cells / T cells** are **transplanted back** into the body of the patient.
- (10) These cells **divide** and proliferate to produce **normal** T cells and/or B cells.

Limitations of gene therapy:

- (11) If the normal allele is **not** successfully **integrated** into the stem cell **genome**, the treatment is **short term** as the **episomal DNA** may be **hydrolysed** and **gene expression will be lost**. Thus the therapy has to be **repeated**.
- (12) If the normal allele is integrated into the host genome at random, **insertional mutagenesis** may occur.

(13) If **viruses** are used to introduce the normal allele, viral **proteins** may trigger an **immune response** in body due to expression of viral genes, and the capsid itself may also trigger immune response.

(14) Virus may regain virulence which cause disease in the patients

[Total: 20]

• END OF PAPER 3 •